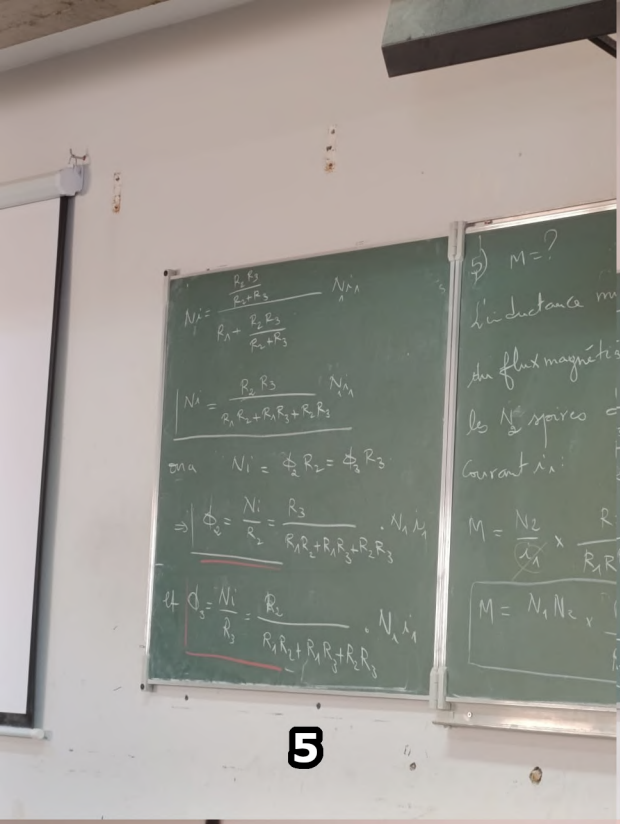
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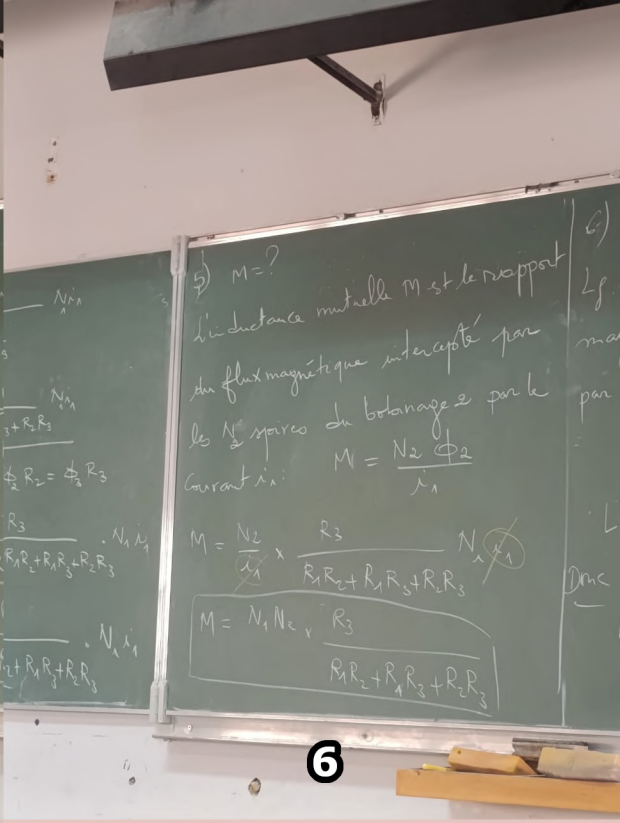
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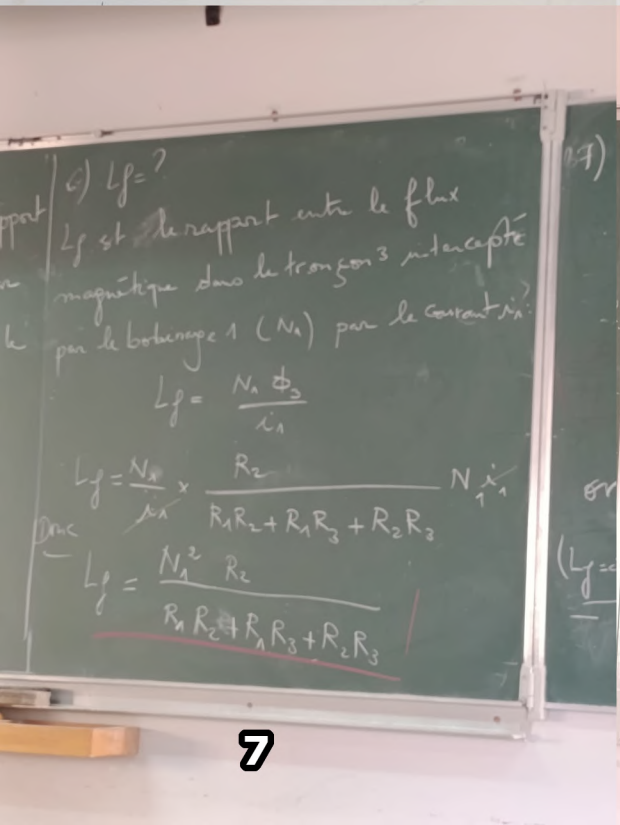
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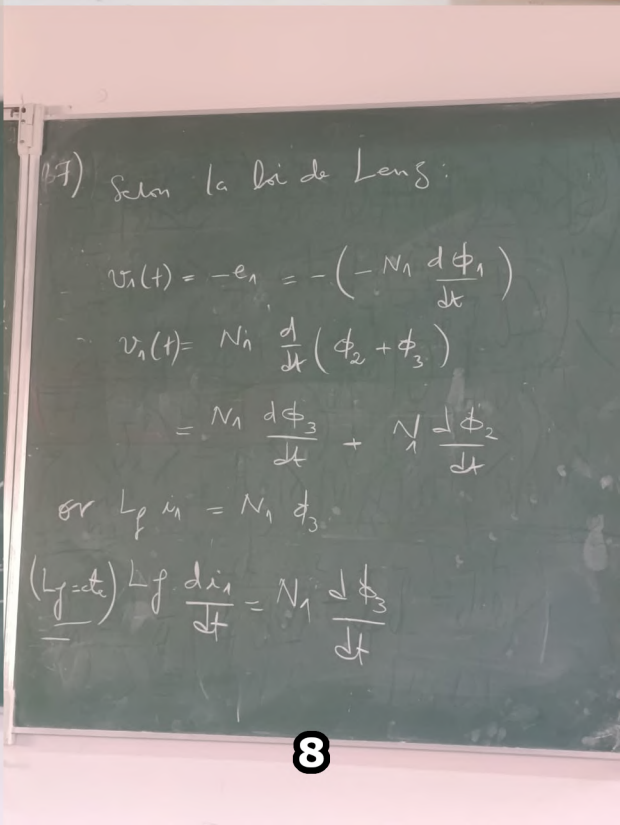
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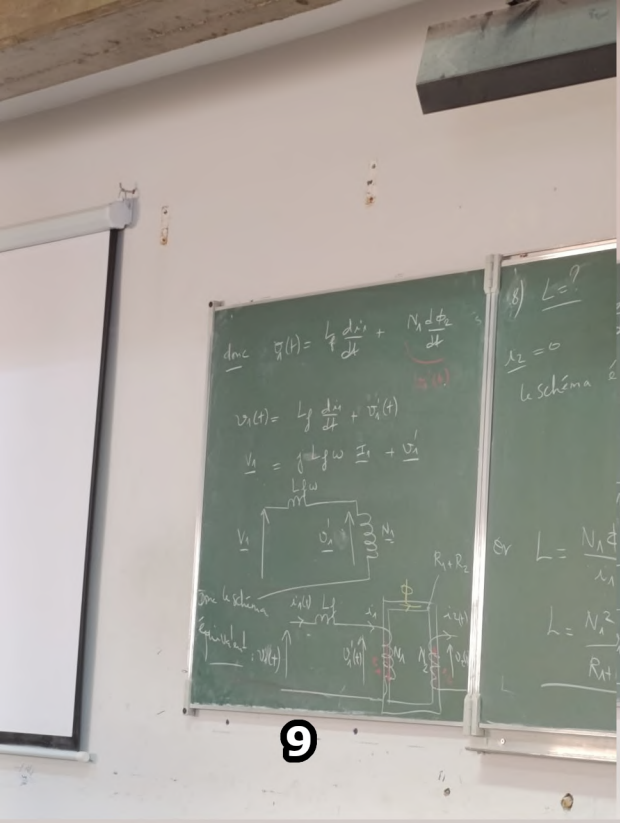
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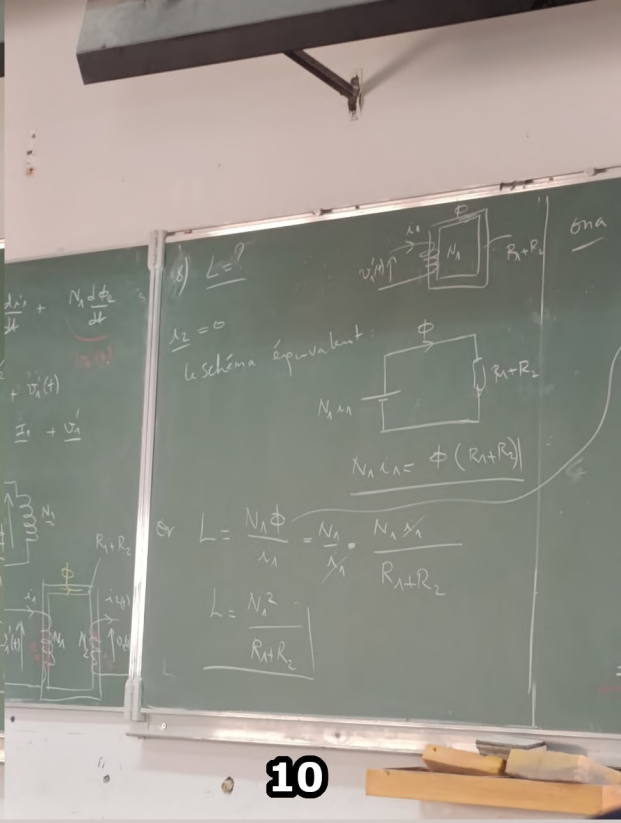


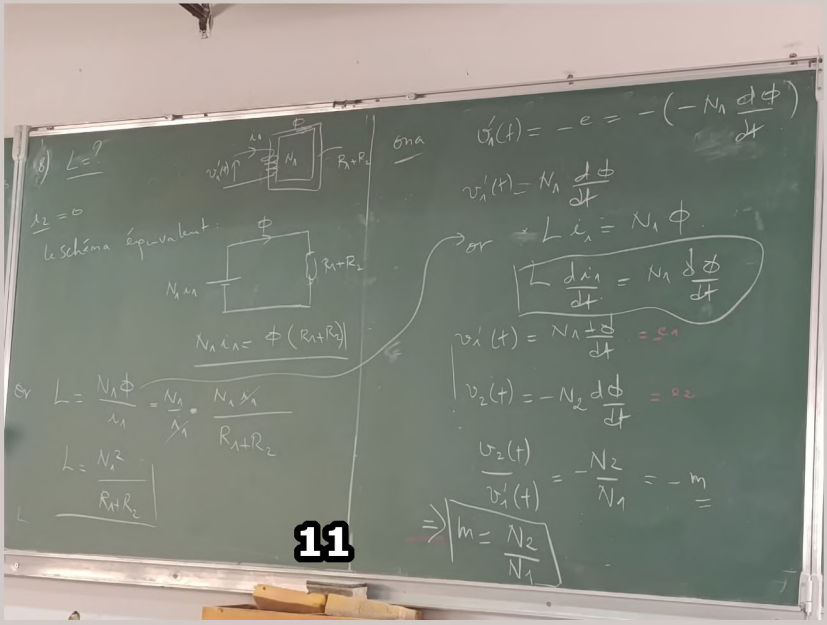
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8


9


10

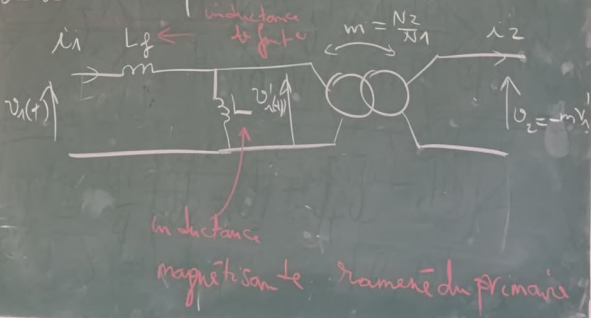

11

12

13

Le rapport permet de représenter le circuit magnétique comme un transformateur parfait de rapport m : a

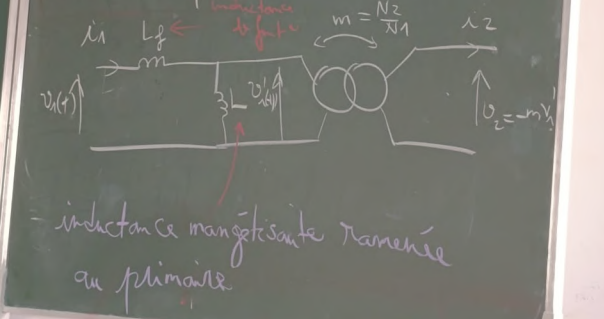
Le schéma équivalent :



14

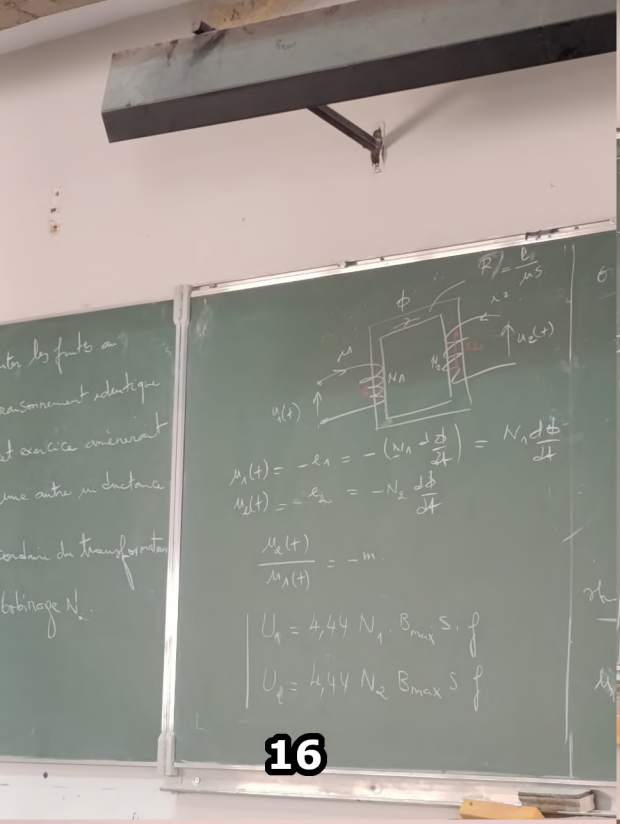
Le rapport permet de représenter le circuit magnétique comme un transformateur parfait de rapport m : a

Le schéma équivalent :

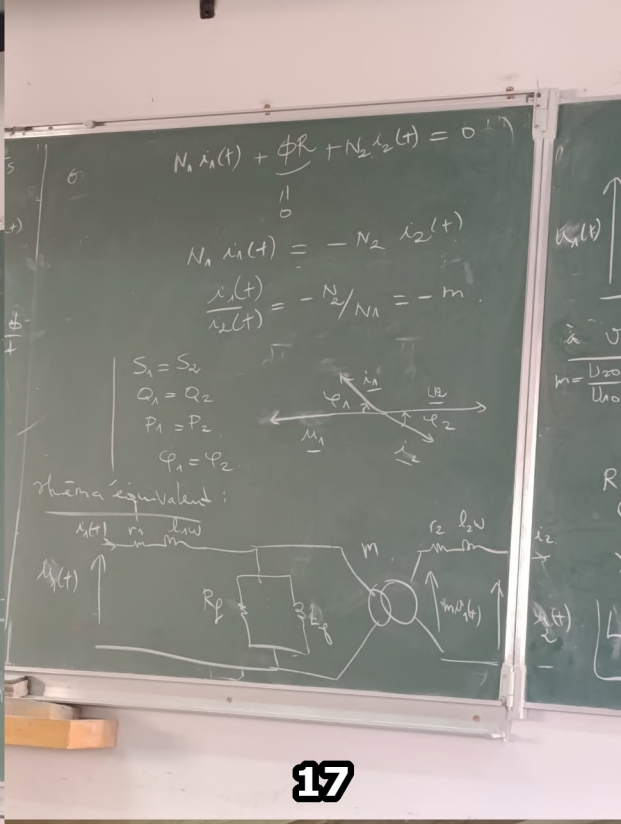


- inductance magnétisante ramenée au primaire

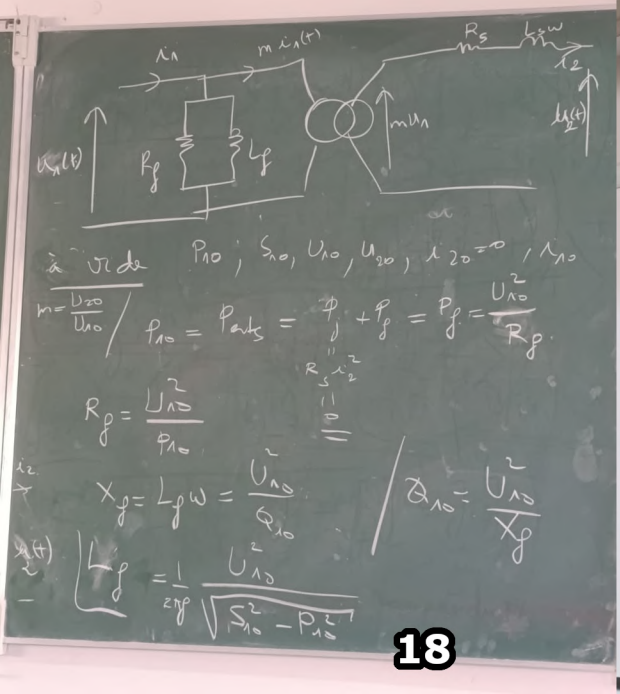
15


$u_1(t) = -e_1 = -\left(N_1 \frac{d\phi}{dt}\right) = N_1 \frac{d\phi}{dt}$
$u_2(t) = -e_2 = -N_2 \frac{d\phi}{dt}$
$\frac{u_2(t)}{u_1(t)} = -m$
$U_1 = 4{,}44\, N_1 \cdot B_{max} \cdot S \cdot f$
$U_2 = 4{,}44\, N_2\, B_{max}\, S\, f$

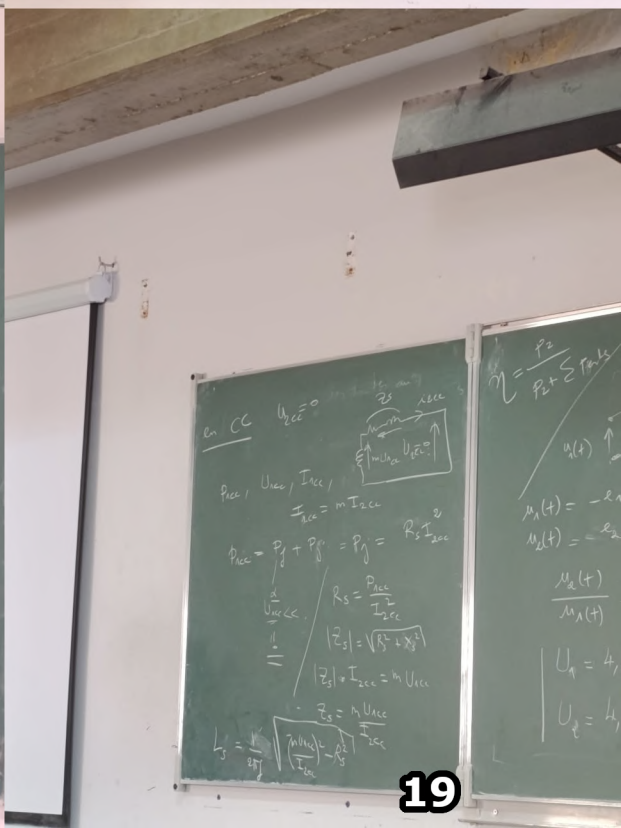
16


$N_1 i_1(t) + \phi R + N_2 i_2(t) = 0$
$N_1 i_1(t) = -N_2 i_2(t)$
$\frac{i_1(t)}{i_2(t)} = -\frac{N_2}{N_1} = -m$
$S_1 = S_2$
$Q_1 = Q_2$
$P_1 = P_2$
$\varphi_1 = \varphi_2$
Schéma équivalent :

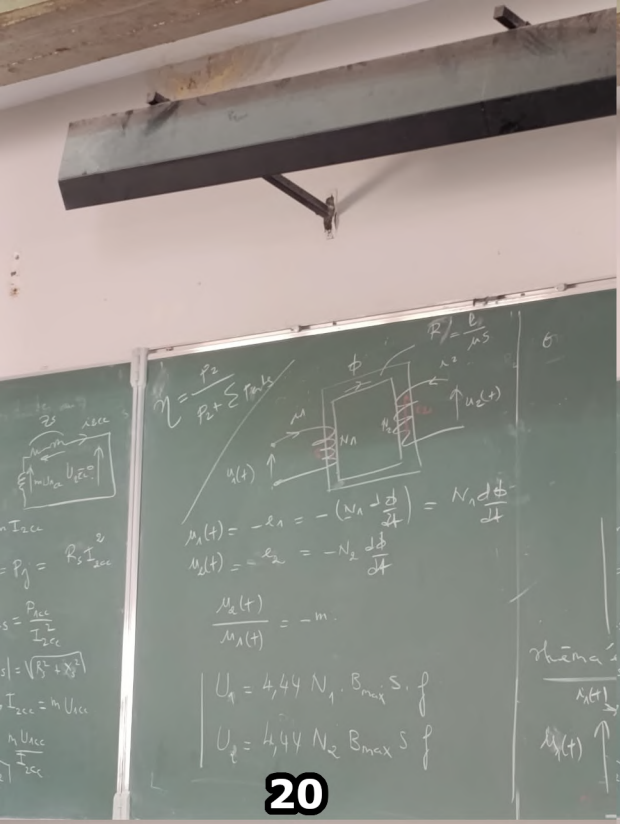
17


$m = \frac{U_{20}}{U_{10}}$
$P_{10} = P_{ferts} = P_j + P_f = P_f = \frac{U_{10}^2}{R_f}$
$R_f = \frac{U_{10}^2}{P_{10}}$
$X_f = L_f \omega = \frac{U_{10}^2}{Q_{10}}$
$Q_{10} = \frac{U_{10}^2}{X_f}$
$L_f = \frac{1}{2\pi f} \frac{U_{10}^2}{\sqrt{S_{10}^2 - P_{10}^2}}$

18


$\eta = \frac{P_2}{P_2 + \sum P_{pertes}}$
$u_1(t) = -e_1$
$u_2(t) = -e_2$
$\frac{u_2(t)}{u_1(t)}$
$U_1 = 4,$
$U_f = 4,$

19



$$\eta = \frac{P_2}{P_2 + \sum \text{pertes}}$$

$$u_1(t) = -e_1 = -\left(N_1 \frac{d\phi}{dt}\right) = N_1 \frac{d\phi}{dt}$$

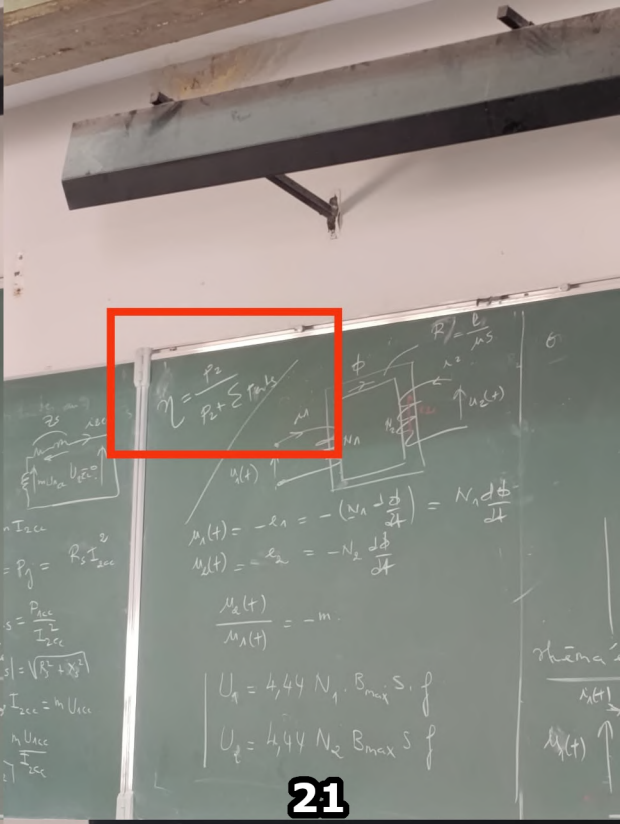
$$u_2(t) = -e_2 = -N_2 \frac{d\phi}{dt}$$

$$\frac{u_2(t)}{u_1(t)} = -m$$

$$U_1 = 4{,}44\, N_1 \cdot B_{max} S \cdot f$$

$$U_2 = 4{,}44\, N_2\, B_{max}\, S\, f$$

20



$$\eta = \frac{P_2}{P_2 + \sum \text{pertes}}$$

$$u_1(t) = -e_1 = -\left(N_1 \frac{d\phi}{dt}\right) = N_1 \frac{d\phi}{dt}$$

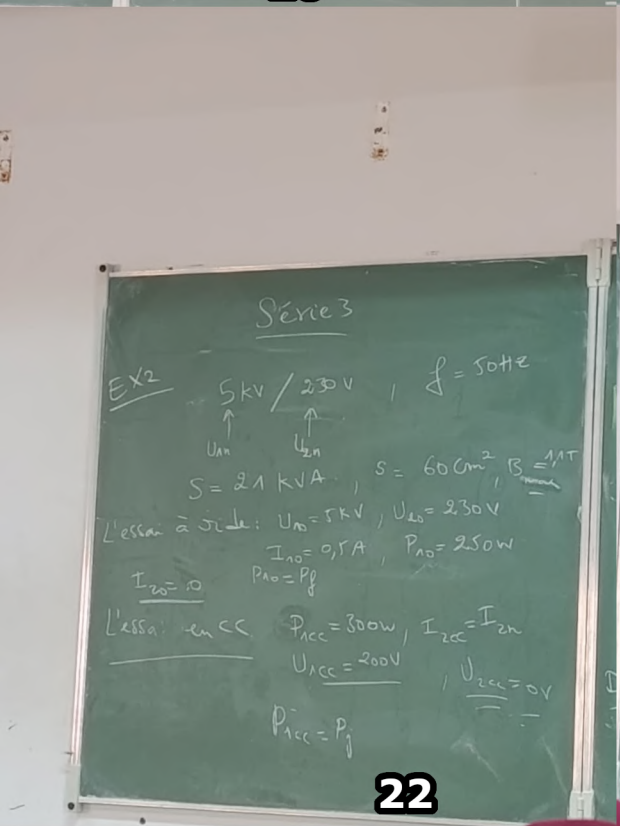
$$u_2(t) = -e_2 = -N_2 \frac{d\phi}{dt}$$

$$\frac{u_2(t)}{u_1(t)} = -m$$

$$U_1 = 4{,}44\, N_1 \cdot B_{max} S \cdot f$$

$$U_2 = 4{,}44\, N_2\, B_{max}\, S\, f$$

21



Série 3

EX2

5 kV / 230 V , $f = 50$ Hz

U_{1n} U_{2n}

$S = 21$ kVA , $S = 60\,cm^2$, $B = 1{,}1$ T

L'essai à vide : $U_{10} = 5$ kV , $U_{20} = 230$ V

$I_{10} = 0{,}5$ A , $P_{10} = 250$ W

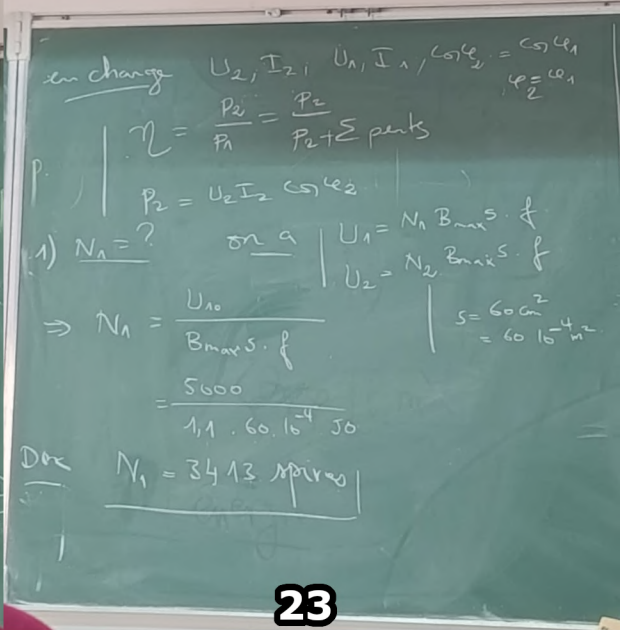
$I_{20} = 0$ $P_{10} = P_f$

L'essai en cc $P_{1cc} = 300$ W , $I_{2cc} = I_{2n}$

$U_{1cc} = 200$ V , $U_{2cc} = 0$ V

$P_{1cc} = P_j$

22



en charge U_2, I_2 ; U_1, I_1 / $\cos\varphi_2 = \cos\varphi_1$, $\varphi_2 = \varphi_1$

$$\eta = \frac{P_2}{P_1} = \frac{P_2}{P_2 + \sum \text{pertes}}$$

$$P_2 = U_2 I_2 \cos\varphi_2$$

1) $N_1 = ?$ on a $U_1 = N_1 B_{max} S \cdot f$, $U_2 = N_2 B_{max} S \cdot f$

$S = 60\,cm^2 = 60\,10^{-4}\,m^2$

$$\Rightarrow N_1 = \frac{U_{10}}{B_{max} S \cdot f} = \frac{5000}{1{,}1 \cdot 60 \cdot 10^{-4} \cdot 50}$$

Donc $N_1 = 3413$ spires

23

2) $m = ?$, $N_2 = ?$

$$m = \frac{U_{20}}{U_{10}} = \frac{230}{5000} = 0,046$$

$$m = \frac{U_{20}}{U_{10}} = \frac{N_2}{N_1} \Rightarrow N_2 = m\, N_1$$

$N_2 = 0,046 \times 3413$

$N_2 = 157$ spires

3) $\cos\varphi_{10} = ?$

$P_{10} = U_{10} \cdot I_{10} \cos\varphi_{10} \Rightarrow \cos\varphi_{10} = \frac{P_{10}}{U_{10} \cdot I_{10}}$

AN $\cos\varphi_{10} = \frac{250}{5000 \cdot 0,5}$

$\cos\varphi_{10} = 0,1$

24

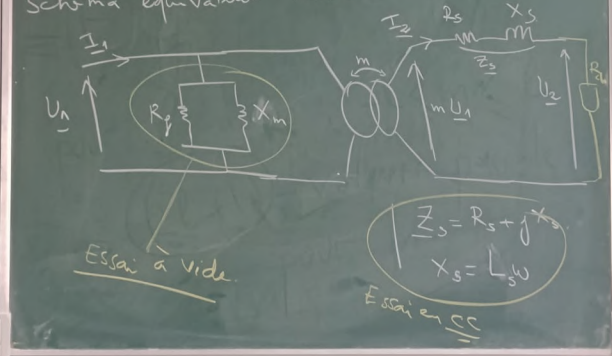
4) $I_{2n} = ?$

on $S = S_1 = S_2 = U_{2n} I_{1n}$ (= U_{20})

$$I_{1n} = \frac{S_2}{U_{20}} = \frac{21.10^3}{230} = 91,30\ A$$

5) $R_s = ?$, $Z_s = ?$, $X_s = ?$

Schéma équivalent ramené au secondaire



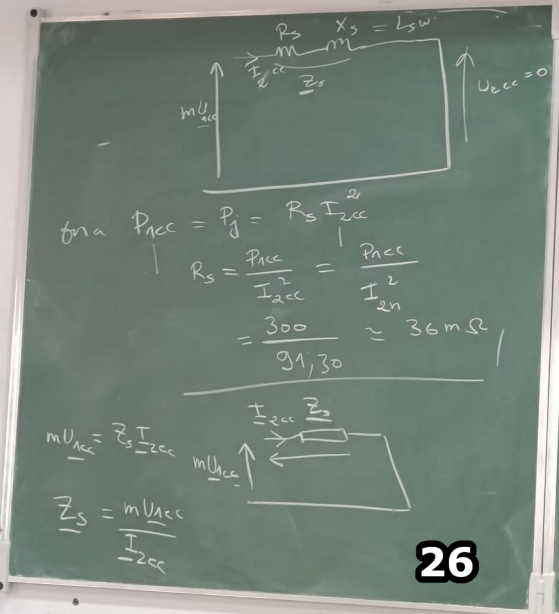
Essai à vide

$\underline{Z}_s = R_s + jX_s$

$X_s = L_s \omega$

Essai en CC

25



donc $P_{1cc} = P_j = R_s I_{2cc}^2$

$$R_s = \frac{P_{1cc}}{I_{2cc}^2} = \frac{P_{1cc}}{I_{2n}^2}$$

$$= \frac{300}{91,30} \simeq 36\ m\Omega$$

$\underline{Z}_s = \frac{m U_{1cc}}{I_{2cc}}$

26

$$|\underline{Z}_s| = \frac{m U_{1cc}}{I_{2cc}} = \frac{0,046 \times 200}{91,30} = 0,1\ \Omega$$

$\underline{Z}_s = R_s + jX_s$

$|\underline{Z}_s|^2 = R_s^2 + X_s^2$

$X_s = \sqrt{Z_s^2 - R_s^2}$

$X_s = \sqrt{\left(\frac{m U_{1cc}}{I_{2cc}}\right)^2 - \left(\frac{P_{1cc}}{I_{2cc}^2}\right)^2}$

$X_s = \sqrt{Z_s^2 - R_s^2}$

$X_s = \sqrt{0,1^2 - (36.10^{-3})^2}$

Donc $X_s \simeq 94\ m\Omega$ $\Rightarrow L_s = \frac{X_s}{\omega} = \frac{X_s}{2\pi f}$

27

6) $\cos\varphi_2 = 0{,}83$, $I_2 = I_{2n}$, $\eta = ?$

$\varphi_2 = 33{,}90°$

$\sin\varphi_2 = 0{,}55$?

ona $\eta = \frac{P_2}{P_1} = \frac{P_2}{P_2 + \sum pertes} = \frac{P_2}{P_2 + P_{fer} + P_j + P_{acc}}$

or $P_2 = U_2 I_2 \cos\varphi_2$

$U_2 = ?$

$U_{20} = m U_1$

$U_2 = U_{20} - \Delta U$

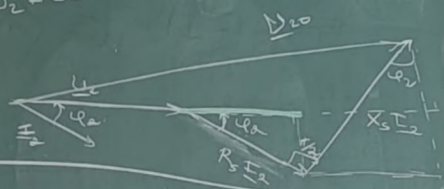
$\Delta U = R_s I_2 + X_s I_2$

28

$U_{20} = U_2 + R_s I_2 + X_s I_2$

ΔU

$U_{20} = U_2 + \Delta U$



avec

$\Delta U = R_s I_2 \cos\varphi_2 + X_s I_2 \sin\varphi_2$

$\Delta U = I_2 (R_s \cos\varphi_2 + X_s \sin\varphi_2)$

$= 91{,}30 (36.10^{-3} \times 0{,}83 + 94{,}10^{-3} \times 0{,}55)$

$\Delta U = 7{,}5\ V$

29

$\Rightarrow U_2 = U_{20} - \Delta U$

$= 230 - 7{,}5$

$U_2 = 222{,}5\ V$

$P_2 = U_2 I_2 \cos\varphi_2$

$= 222{,}5 \times 91{,}30 \times 0{,}83$

$P_2 = 16{,}86\ kW$

$P_1 = P_2 + \sum pertes = P_2 + P_{fer} + P_{acc}$

$= 16{,}86.10^3 + 250 + 300 = 17{,}41\ kW$

$\eta = P_2 / P_1 = 16{,}86 / 17{,}41 = 0{,}9684 = 96{,}84\ \%$

30

Ex3 $S = 2200\ VA$ | $V_{1n} = 220\ V$ | $V_{2n} = 127\ V$; $\eta = 0{,}95$

1) $I_{1n} = ?$ $I_{1n} = S / V_{1n} = \frac{2200}{220} = 10\ A$

2) $I_{2n} = ?$ $I_{2n} = S / V_{2n} = \frac{2200}{127} = 17{,}32\ A$

3) $\cos\varphi_2 = 0{,}8$, $I_2 = I_{2n}$, $V_2 = V_{2n}$, $\sum pertes = ?$

ona $\eta = \frac{P_2}{P_2 + \sum pertes}$

$\sum pertes = \frac{P_2}{\eta} - P_2 = P_2 \left(\frac{1}{\eta} - 1\right)$

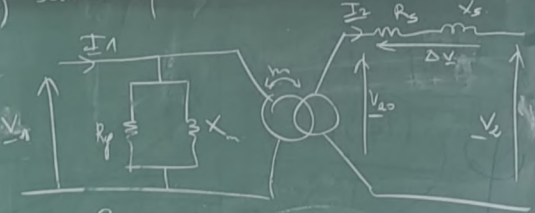
or $P_2 = V_{2n} I_{2n} \cos\varphi_2 = 127 \times 17{,}32 \times 0{,}8$

$P_2 = 1759{,}71\ W$

31

Σ pertes = 92,63 W

4) Schéma équivalent ramené au secondaire :



5) R_f = ? & R_s = ?

Σ pertes = $P_f + P_j = 2P_f = 2P_j$

$P_j = P_f = \frac{\Sigma \text{ pertes}}{2} = 46,315$ W

32

*) $P_f = \frac{V_1^2}{R_f} \Rightarrow R_f = \frac{V_1^2}{P_f} = \frac{(220)^2}{46,315}$

$R_f = 1045\ \Omega$

*) $P_j = R_s I_{2n}^2 \Rightarrow R_s = \frac{P_j}{I_{2n}^2} = \frac{46,315}{(17,32)^2}$

$R_s = 0,154\ \Omega$

c) $V_{20} = 133$ V, m = ?, ΔV_2 = ?, X_s = ?

on a $m = \frac{V_{20}}{V_{1n}} = \frac{133}{220} = 0,604$

33

on a $V_{20} = \Delta V + V_2$

$\Delta V = V_{20} - V_2$ ($V_2 = V_{2n}$)

$= 133 - 127$

$\Delta V = 6$ V

on a $\Delta V = R_s I_2 + X_s I_2$

$\Delta V = R_s I_2 \cos\varphi_2 + X_s I_2 \sin\varphi_2$

$X_s = \frac{\Delta V - R_s I_2 \cos\varphi_2}{I_2 \sin\varphi_2}$

$\cos\varphi_2 = 0,8 \rightarrow \sin\varphi_2 = 0,6$

34

$X_s = \frac{6 - 0,154 \times 0,8 \times 17,32}{17,32 \times 0,6}$

$X_s = 0,37\ \Omega$

$X_s = L_s \omega \Rightarrow L_s = \frac{X_s}{2\pi f} = \frac{0,37}{100\pi}$

$L_s = 1,184$ mH

f) V_2 = ?, $I_2 = \frac{I_{2n}}{2}$, $\cos\varphi_2 = 0,8$

on a $V_2 = V_{20} - \Delta V$

$= V_{20} - I_2 (R_s \cos\varphi_2 + X_s \sin\varphi_2)$

($I_2 = \frac{I_{2n}}{2}$)

$V_2 = 133 - \frac{17,32}{2}(0,154 \times 0,8 + 0,37 \times 0,6)$

35

$V_2 = 130,01\ V$

8) $\eta = ?$

$\eta = \frac{P_2}{P_2 + \sum pertes}$

$P_2 = V_2 \times I_2 \times \cos\varphi = 130,01 \times \frac{17,32}{2} \times 0,8$

$I_2 = \frac{I_{2n}}{2}$

$P_2 = 900,71\ W$

$\sum pertes = P_j + P_f = R_s \left(\frac{I_{2n}}{2}\right)^2 +$

$= 11,15\ W$

$46,31\ TW$

$= V1^2$

R_f

$= Cte$

$=$

36

$\Rightarrow \sum pertes = 57,86\ W$

$\eta = \frac{P_2}{P_2 + \sum pertes}$

$= \frac{900,71}{900,71 + 57,86}$

$= 0,9396 \simeq 0,94$

$\eta = 94\ \%$

37